Immunodeficiency 82 With Systemic Inflammation (IMD82) — Comprehensive Disease Characterization Report

Target Disease: Immunodeficiency 82 With Systemic Inflammation **OMIM:** #619381 · **MONDO:** MONDO:0030308 · **Causal gene:** *SYK* (HGNC:11491) **Category:** Mendelian (monogenic inborn error of immunity)

Summary

Immunodeficiency 82 with systemic inflammation (IMD82) is an ultra-rare, autosomal dominant inborn error of immunity (IEI) caused by germline heterozygous gain-of-function (GOF) missense variants in *SYK* (spleen tyrosine kinase). The disease was defined in 2021 by Wang and colleagues, who identified damaging monoallelic *SYK* variants in six patients presenting with a distinctive triad: immune deficiency, multi-organ inflammatory disease (colitis, arthritis, dermatitis), and diffuse large B-cell lymphoma (DLBCL) (PMID: 33782605). The variants increase SYK phosphorylation and enhance downstream signaling, establishing a gain-of-function mechanism.

Mechanistically, SYK is normally held in an **autoinhibited conformation** in which its tandem SH2 (tSH2) domain and interdomain linkers constrain the kinase domain. Binding of phosphorylated immunoreceptor tyrosine-based activation motifs (pITAMs) plus autophosphorylation switches SYK to its active state (PMID: 23154170). IMD82 variants bias SYK toward the active conformation, producing **constitutive ITAM-based signaling** through PLCγ2, PI3K–AKT, MAPK/ERK, and NF-κB. This single hyperactive-kinase lesion simultaneously drives the inflammatory arm (excess cytokine/immune-cell activation) and the malignant arm (sustained pro-survival BCR/PI3K signaling that predisposes B cells to lymphoma).

Because the driver is a **druggable, hematopoietically restricted kinase**, IMD82 is potentially treatable. A knock-in *Syk*-Ser544Tyr mouse (modeling the patient variant p.Ser550Tyr) recapitulated key disease features that could be partially corrected either with a SYK inhibitor or with transplantation of wild-type bone marrow (PMID: 33782605). Fostamatinib, an FDA-

approved oral SYK inhibitor (approved for chronic immune thrombocytopenia), provides a clinically available agent whose mechanism directly matches the disease lesion. Diagnosis relies on next-generation sequencing (whole-exome/whole-genome sequencing or IEI gene panels) with confirmatory functional testing of elevated SYK phosphorylation/downstream signaling. Only ~6 patients have been reported worldwide, so much of the clinical natural history remains to be defined.

Key Findings

Finding 1 — IMD82 is caused by monoallelic gain-of-function variants in *SYK*

Wang et al. (2021, *Nature Genetics*) identified damaging **monoallelic (heterozygous) *SYK* variants** in six patients. Functional assays demonstrated that the variants **increased SYK phosphorylation and enhanced downstream signaling**, formally establishing a gain-of-function (rather than loss-of-function) mechanism and an autosomal dominant mode of inheritance. This is the defining genetic feature of the disease and is catalogued as OMIM #619381 / MONDO:0030308.

> *"We identified damaging monoallelic SYK variants in six patients with immune deficiency, multi-organ inflammatory disease such as colitis, arthritis and dermatitis, and diffuse large B cell lymphomas. The SYK variants increased phosphorylation and enhanced downstream signaling, indicating gain of function."* — PMID: 33782605

This finding is notable because most *SYK*-related disease hypotheses historically concerned loss of function; IMD82 is a rare instance where a kinase gain-of-function drives a combined immunodeficiency-plus-autoinflammation syndrome.

Finding 2 — SYK is normally autoinhibited; GOF variants disrupt the activation switch

Crystal structures of full-length SYK reveal an **autoinhibited conformation** in which the tandem SH2 (tSH2) domain and interdomain linkers restrain kinase activity. Binding of phosphorylated ITAMs to the tSH2 domain, together with autophosphorylation of interdomain-A tyrosines (e.g., Y348/Y352), switches SYK to the active conformation and strongly stimulates in vitro autophosphorylation (PMID: 23154170). IMD82 patient variants (e.g., **p.Ser550Tyr**, and residues in the interdomain-A/kinase regions) constitutively bias this switch toward the active state, producing elevated basal phosphorylation and downstream signaling.

> *"Here, we present the first crystal structures of full-length Syk (fl-Syk) as wild type and as Y348F,Y352F mutant forms in complex with AMP-PNP revealing an autoinhibited conformation."* — PMID: 23154170
>
> *"The functional relevance of pITAM binding to fl-Syk was confirmed by a strong stimulation of in vitro autophosphorylation."* — PMID: 23154170

This provides a clear structural rationale for the disease: the GOF variants relieve the molecular brake that normally couples SYK activation to receptor engagement.

Finding 3 — Constitutive SYK signaling links the inflammatory and lymphoma-predisposition arms

SYK is the proximal kinase that transduces signals from the B-cell receptor (BCR) and Fc/C-type-lectin receptors into downstream **PI3K–AKT, PLCγ2, MAPK/ERK, and NF-κB** cascades (PMID: 35398488; PMID: 38503806). Sustained/tonic PI3K-pathway signaling downstream of SYK promotes B-cell **survival** rather than negative selection, and high SYK/BCR signaling activity predicts aggressive B-cell lymphoma behavior. This mechanistically unifies the two seemingly disparate arms of IMD82 — chronic multi-organ inflammation and predisposition to DLBCL — under a single constitutively active kinase.

> *"ZAP70 diverts SYK from activation of NFAT towards tonic PI3K-signaling, which promotes survival instead of cell death"* — PMID: 35398488

In mantle cell lymphoma, higher BCR/SYK signaling responses correlate with shorter progression-free and overall survival, underscoring that SYK signaling intensity is a determinant of malignant aggressiveness (PMID: 38503806).

Finding 4 — Clinical phenotype: pediatric-onset immunodeficiency + multi-organ autoinflammation + DLBCL

In the defining cohort of six patients, the phenotype combined **immune deficiency** with **multi-organ inflammatory disease** — colitis (HP:0002583), arthritis (HP:0001369), and dermatitis (HP:0011123) — plus **diffuse large B-cell lymphoma** (HP:0004297) (PMID: 33782605). Onset is typically in childhood, frequently as a very-early-onset inflammatory-bowel-disease-like presentation. IMD82/SYK is incorporated into the formal IUIS Inborn Errors of Immunity classification framework used for diagnosis (PMID: 41608114).

> *"immune deficiency, multi-organ inflammatory disease such as colitis, arthritis and dermatitis, and diffuse large B cell lymphomas"* — PMID: 33782605
>
> *"This report provides an updated classification of inborn errors of immunity (IEIs) involving 508 different genes and 17 phenocopies."* — PMID: 41608114

Finding 5 — Ultra-rare autosomal dominant disorder diagnosed by exome/genome sequencing

Only ~6 patients were reported in the defining publication, and no population prevalence/incidence estimates exist (ultra-rare). Inheritance is autosomal dominant (monoallelic/heterozygous variants), consistent with both de novo and inherited transmission. Diagnosis relies on next-generation sequencing (WES/WGS or IEI gene panels), as used across monogenic VEO-IBD and IEI cohorts, with confirmatory functional assays demonstrating elevated SYK phosphorylation/downstream signaling (PMID: 33782605).

> *"We identified damaging monoallelic SYK variants in six patients"* — PMID: 33782605

Finding 6 — The disease is potentially treatable with SYK inhibition or bone-marrow transplant

A knock-in **Syk-Ser544Tyr mouse** (modeling patient variant p.Ser550Tyr) recapitulated aspects of the human disease that could be **partially treated with a SYK inhibitor** or with **transplantation of wild-type bone marrow** (PMID: 33782605). Fostamatinib is an FDA-approved oral SYK inhibitor (approved for chronic ITP), making targeted pharmacotherapy immediately plausible.

> *"A knock-in (SYK-Ser544Tyr) mouse model of a patient variant (p.Ser550Tyr) recapitulated aspects of the human disease that could be partially treated with a SYK inhibitor or transplantation of bone marrow from wild-type mice."* — PMID: 33782605

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **germline heterozygous gain-of-function missense variant in *SYK*** (e.g., p.Ser550Tyr) arises de novo or is inherited (autosomal dominant). — *demonstrated* (PMID: 33782605)
2. The variant **disrupts SYK's autoinhibited conformation** — normally maintained by the tSH2 domain and interdomain-A linkers — biasing the kinase toward its active state. — *inferred from structure* (PMID: 23154170)
3. This **relieves the requirement for pITAM engagement/autophosphorylation**, so SYK exhibits **elevated basal (constitutive) kinase activity** and enhanced phosphorylation independent of full receptor input. — *demonstrated (increased pSYK)* (PMID: 33782605)
4. Constitutive SYK activity **hyper-transduces ITAM-coupled receptor signals** (BCR, Fc receptors, C-type lectin receptors) into downstream cascades: **PLCγ2, PI3K–AKT, MAPK/ERK, and NF-κB**. — *demonstrated/inferred* (PMID: 35398488)
5. The mechanism then **branches**:
6. **Branch A (autoinflammation/immune dysregulation):** Excess NF-κB/MAPK-driven cytokine production and immune-cell activation in myeloid and lymphoid compartments → **multi-organ inflammation** (colitis, arthritis, dermatitis). Paradoxical **immune deficiency** co-exists, reflecting dysregulated rather than simply amplified immunity. — *demonstrated clinically* (PMID: 33782605)
7. **Branch B (lymphomagenesis):** Sustained tonic **PI3K–AKT survival signaling** downstream of SYK promotes B-cell survival over negative selection, permitting accumulation of transformation-prone clones → **diffuse large B-cell lymphoma**. — *inferred from lymphoma biology* (PMID: 35398488; PMID: 38503806)
8. Because the lesion is a **hematopoietically restricted, druggable kinase**, **SYK inhibition** (pharmacologic) or **replacement of the mutant hematopoietic compartment** (WT bone-marrow transplant) partially reverses disease. — *demonstrated in mouse* (PMID: 33782605)

Schematic

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SYK GOF variant (e.g. p.Ser550Tyr)
               │  disrupts
               ▼
Autoinhibited SYK  ──X──►  Constitutively ACTIVE SYK
 (tSH2 brake)                       │ hyper-signaling
                                    ▼
      ┌──────────── ITAM-coupled receptor cascades ────────────┐
      │      PLCγ2 · PI3K–AKT · MAPK/ERK · NF-κB                 │
      └──────────────┬───────────────────────┬─────────────────┘
                     │                       │
           BRANCH A (inflammation)     BRANCH B (survival)
           cytokines, immune            tonic PI3K–AKT →
           dysregulation                B-cell survival
                     │                       │
                     ▼                       ▼
      Colitis, arthritis,           Diffuse large B-cell
      dermatitis + immune           lymphoma (DLBCL)
      deficiency
                     │                       │
                     └──── TREATABLE via ────┘
             SYK inhibitor (fostamatinib) │ WT BMT/HSCT
```

Upstream vs downstream

- **Upstream (initiating):** the *SYK* GOF variant and loss of autoinhibition.
- **Midstream (amplifying):** constitutive PLCγ2/PI3K–AKT/MAPK/NF-κB signaling.
- **Downstream (manifesting):** organ inflammation and B-cell lymphomagenesis.

Suggested ontology terms

- **Gene/protein:** *SYK* (HGNC:11491); GO:0004715 (non-membrane spanning protein tyrosine kinase activity); GO:0050853 (B cell receptor signaling pathway); GO:0038094 (Fc receptor signaling); GO:0002250 (adaptive immune response); GO:0006954 (inflammatory response); GO:0043066 (negative regulation of apoptotic process).
- **Cell types (CL):** CL:0000236 (B cell), CL:0000235 (macrophage), CL:0000775 (neutrophil), CL:0000542 (lymphocyte).
- **Subcellular (GO CC):** GO:0005829 (cytosol), GO:0005886 (plasma membrane / receptor-proximal).

- **Anatomy (UBERON):** UBERON:0000160 (intestine) — colitis; joint/synovium — arthritis; UBERON:0002097 (skin) — dermatitis; UBERON:0002509 (lymphoid tissue) — lymphoma.
- **Chemical (CHEBI):** fostamatinib / R788 (SYK inhibitor prodrug); ATP-competitive kinase inhibition.
- **Disease:** MONDO:0030308; **Phenotypes (HPO):** HP:0002583 (colitis), HP:0001369 (arthritis), HP:0011123 (inflammatory abnormality of the skin / dermatitis), HP:0004297 (neoplasm of the immune system / lymphoma), HP:0002715 (abnormality of the immune system / immunodeficiency).

Section-by-Section Report

1. Disease Information

IMD82 is a Mendelian inborn error of immunity in which a single hyperactive kinase produces a combined phenotype of immunodeficiency, systemic autoinflammation, and B-cell lymphoma predisposition. - **Key identifiers:** OMIM #619381; MONDO:0030308; causal gene *SYK* (HGNC:11491). Orphanet, ICD-10/ICD-11, and MeSH-specific codes are not clearly established for this ultra-rare 2021-defined entity; it maps within IEI/combined immunodeficiency-with-associated-features categories. - **Synonyms:** "Immunodeficiency 82 with systemic inflammation"; "SYK gain-of-function disease"; "SYK-GOF immune dysregulation." - **Information source:** Aggregated disease-level resources (OMIM, IUIS classification) built from an individual-patient case series (n≈6) rather than large EHR datasets.

2. Etiology

- **Causal factor:** Genetic — germline heterozygous GOF missense variants in *SYK* (monogenic, autosomal dominant) (PMID: 33782605).
- **Genetic risk factors:** The causal *SYK* variant itself is necessary and sufficient. Modifier genes are not defined given the tiny cohort. ZAP70 (SYK's paralog) shapes SYK signaling output in principle (PMID: 35398488).
- **Environmental risk factors:** No established environmental triggers; inflammatory flares plausibly modulated by microbial/antigenic exposure via ITAM-coupled receptors (inferred, not demonstrated).

- **Protective factors:** None defined. By mechanistic logic, reduced SYK signaling (pharmacologic inhibition) is protective against manifestations.
- **Gene–environment interactions:** Not characterized; the ITAM-receptor biology implies antigen/microbe exposure could amplify constitutive signaling.

3. Phenotypes

Phenotype	Type	HPO	Onset	Frequency (n≈6 cohort)
Colitis / IBD-like enteropathy	Clinical sign	HP:0002583	Childhood/very-early-onset	Core feature
Arthritis	Clinical sign	HP:0001369	Childhood	Core feature
Dermatitis	Physical manifestation	HP:0011123	Childhood	Core feature
Immunodeficiency (infection susceptibility)	Clinical/lab	HP:0002715	Childhood	Core feature
Diffuse large B-cell lymphoma	Neoplasm	HP:0004297	Childhood–young adult	Reported in cohort

Severity is variable and can be severe/refractory; progression includes both chronic inflammation and episodic flares plus a serious oncologic complication (DLBCL). Quality-of-life impact is substantial (chronic multi-organ inflammation, malignancy risk, treatment burden), though formal EQ-5D/SF-36 data are unavailable for this ultra-rare disease (PMID: 33782605).

4. Genetic / Molecular Information

- **Causal gene:** *SYK* (spleen tyrosine kinase), chromosome 9q22, HGNC:11491; encodes a cytoplasmic non-receptor tyrosine kinase with tandem SH2 domains + kinase domain.
- **Variant classification/type:** Pathogenic/likely pathogenic **missense** variants (e.g., **p.Ser550Tyr**) affecting the activation switch (interdomain-A/kinase regions) (PMID: 33782605).
- **Functional consequence: Gain of function** — increased phosphorylation and enhanced downstream signaling.
- **Allele frequency:** Effectively absent from population databases (gnomAD) given pathogenicity and rarity.

- **Origin:** Germline (monoallelic); both de novo and inherited transmission consistent with AD inheritance. (Note: a distinct *somatic* ETV6–SYK fusion drives chronic eosinophilic leukemia — a separate entity, not IMD82 — PMID: 42005114.)
- **Modifier genes / epigenetics / chromosomal abnormalities:** Not defined for this disorder.

5. Environmental Information

No established environmental, lifestyle, or infectious causes. The disease is monogenic. Because SYK sits downstream of pattern-recognition and Fc/C-type-lectin receptors, microbial and antigenic exposures are plausible amplifiers of inflammation but are not demonstrated drivers.

6. Mechanism / Pathophysiology

Presented above as the ordered causal chain and schematic. In brief: a *SYK* GOF variant disrupts kinase autoinhibition → constitutive ITAM-coupled signaling (PLCγ2, PI3K–AKT, MAPK/ERK, NF-κB) → branches into (A) multi-organ autoinflammation with paradoxical immunodeficiency and (B) tonic PI3K–AKT survival signaling predisposing to DLBCL. Key molecular pathways: **B-cell receptor signaling (Reactome/KEGG), Fc receptor signaling, PI3K–AKT, MAPK, NF-κB**. Cellular processes: inflammation, dysregulated lymphocyte/myeloid activation, and suppressed apoptosis/enhanced survival of B cells. Immune involvement spans both immunodeficiency and chronic inflammation — a hallmark of immune-dysregulation IEIs.

7. Anatomical Structures Affected

- **Primary organs/systems:** Gastrointestinal tract (colon — colitis; UBERON:0000160), joints (arthritis; synovium), skin (dermatitis; UBERON:0002097), and the immune/lymphoid system (UBERON:0002509).
- **Secondary/complication:** Lymphoid tissues affected by DLBCL; systemic effects of chronic inflammation.
- **Cell populations:** B lymphocytes (CL:0000236) central; macrophages (CL:0000235), neutrophils (CL:0000775), and other ITAM-receptor-bearing hematopoietic cells contribute.
- **Subcellular:** Cytosolic/plasma-membrane-proximal signaling compartments (GO:0005829, GO:0005886).
- **Laterality:** Systemic/bilateral (not a lateralized disease).

8. Temporal Development

- **Onset:** Pediatric/childhood, often very-early-onset (VEO-IBD-like presentation); course is chronic with episodic inflammatory flares.
- **Progression:** Variable; chronic multi-organ inflammation punctuated by the serious late complication of DLBCL.
- **Critical windows:** Early SYK-targeted intervention is mechanistically attractive to blunt both inflammation and lymphoma risk (inferred from mouse rescue data, PMID: 33782605).

9. Inheritance and Population

- **Epidemiology:** Ultra-rare; ~6 reported patients; no prevalence/incidence estimates.
- **Inheritance:** Autosomal dominant (monoallelic/heterozygous). Penetrance/expressivity not quantifiable in so small a cohort; both de novo and inherited variants are consistent with the genetics.
- **Founder effects/consanguinity/carrier frequency:** Not applicable (dominant, ultra-rare).
- **Demographics:** No described ethnic, geographic, or sex predilection.

10. Diagnostics

- **Recommended approach:** Next-generation sequencing — **whole-exome or whole-genome sequencing**, or **IEI/VEO-IBD gene panels** including *SYK* — is the diagnostic backbone (PMID: 33782605; cohorts in PMID: 42494428, PMID: 42724325).
- **Confirmatory functional testing:** Phospho-specific assays demonstrating **elevated basal/inducible SYK phosphorylation (pSYK) and enhanced downstream signaling** distinguish GOF variants from benign or LOF variants — a key step because *SYK* variant interpretation requires functional context.
- **Supportive workup:** Immunophenotyping/immunoglobulin levels (immunodeficiency evaluation), inflammatory markers, endoscopy/biopsy for colitis, and oncologic staging if lymphoma is suspected.
- **Differential diagnosis:** Other monogenic immune-dysregulation/VEO-IBD disorders (e.g., CTLA4 haploinsufficiency — PMID: 42577265), other IEIs with autoimmunity/lymphoproliferation (PMID: 42625995). Functional pSYK testing plus the specific *SYK* variant distinguishes IMD82.

11. Outcome / Prognosis

No formal survival statistics exist for this ultra-rare entity. Prognosis is shaped by (a) severity/refractoriness of multi-organ inflammation and (b) the serious complication of DLBCL, which historically carries substantial mortality. The identification of a **druggable, hematopoietically restricted target** materially improves the outlook, since SYK inhibition or HSCT can address the root cause (mouse-model rescue, PMID: 33782605). Prognostic biomarkers plausibly include SYK/BCR signaling intensity, given its correlation with aggressive B-cell lymphoma behavior (PMID: 38503806).

12. Treatment

- **Targeted pharmacotherapy (NCIT: spleen tyrosine kinase inhibitor): Fostamatinib** (R788), an FDA-approved oral SYK inhibitor (approved for chronic ITP), mechanistically matches the lesion; mouse data show a SYK inhibitor partially treats the disease (PMID: 33782605). Fostamatinib's efficacy in other SYK-driven immune disorders (ITP, warm AIHA, TRAF3-deficient autoimmunity) supports the class rationale (PMID: 40903253; PMID: 42026764). **Safety note:** fostamatinib carries potential vascular/thrombotic and off-target concerns (e.g., a reported aortic-dissection case, PMID: 40962713).
- **Cell therapy (NCIT: hematopoietic stem cell transplantation): Allogeneic HSCT / wild-type bone-marrow transplant** is curative in the mouse model and is a rational option because the disease is hematopoietically restricted (PMID: 33782605).
- **Supportive/anti-inflammatory therapy:** Standard IEI/IBD management (immunoglobulin replacement for immunodeficiency; anti-inflammatory/biologic/JAK-inhibitor strategies for colitis/arthritis/dermatitis, extrapolated from VEO-IBD practice, PMID: 42499735).
- **Lymphoma therapy:** Standard DLBCL protocols if malignancy develops; SYK/BCR-pathway inhibitors (SYK, BTK, PI3K) are active in B-cell malignancies (PMID: 38660880; PMID: 35296646).
- **Personalized medicine:** IMD82 is a paradigm for **genotype-guided targeted therapy** — the causal kinase is the drug target.

13. Prevention

- **Primary prevention:** Not applicable (monogenic); **genetic counseling** for affected families given autosomal dominant inheritance.
- **Secondary prevention:** Cascade genetic testing of relatives; surveillance for lymphoma and inflammatory complications in carriers.

- **Tertiary prevention:** Early SYK-targeted therapy or HSCT to limit organ damage and potentially reduce lymphoma risk (inferred).
- **Prenatal/preimplantation testing:** Available in principle once the familial variant is known.

14. Other Species / Natural Disease

- **Model taxonomy:** *Mus musculus* (NCBI Taxon 10090) via the knock-in Syk-Ser544Tyr allele (PMID: 33782605).
- **Orthologs:** *Syk* is conserved across vertebrates, including jawless fish (lamprey Lj-Syk), with conserved tandem-SH2 and kinase domains (PMID: 25682127).
- **Natural disease in other species:** No naturally occurring animal counterpart of IMD82 is documented (OMIA); the disease is defined in humans with an engineered mouse model.

15. Model Organisms

- **Principal model: Knock-in mouse Syk-Ser544Tyr**, engineered to mirror the human p.Ser550Tyr GOF variant. It **recapitulated aspects of the human disease** and, critically, was **partially rescued by a SYK inhibitor or by wild-type bone-marrow transplantation** (PMID: 33782605) — providing both face validity and therapeutic proof-of-concept.
- **Complementary tools:** Chemical-genetic (analog-sensitive) SYK mouse systems and myeloid-specific SYK knockouts exist for dissecting SYK-dependent signaling (PMID: 31356155), and B-cell TRAF3-deficient mice model SYK-driven autoimmunity/lymphoma responsive to fostamatinib (PMID: 42026764).
- **Limitations:** Partial (not complete) recapitulation/rescue; small human cohort limits cross-validation; models emphasize signaling and inflammation more than the full lymphoma trajectory.

Evidence Base

PMID	Title (abbrev.)	Role in this report
33782605	*Gain-of-function variants in SYK cause immune dysregulation and systemic inflammation* (Nat Genet 2021)	**Defining paper.** Establishes monoallelic SYK GOF, clinical triad, mouse model, and therapeutic rescue (SYK inhibitor / WT BMT).
23154170	*Structural and biophysical characterization of the Syk activation switch*	Structural basis: full-length SYK autoinhibition and the pITAM/autophosphorylation activation switch disrupted by GOF variants.
35398488	*SYK and ZAP70 kinases in autoimmunity and lymphoid malignancies*	Mechanistic link: SYK-driven tonic PI3K signaling promotes survival over cell death — basis for lymphoma predisposition.
38503806	*BCR signaling activity identifies higher-risk MCL patients*	High SYK/BCR signaling predicts aggressive B-cell lymphoma behavior; supports Branch B.
41608114	*Human inborn errors of immunity: 2024 IUIS update*	Situates IMD82/SYK within the formal IEI classification used for diagnosis.
42026764	*Syk inhibition limits autoimmunity in B-cell TRAF3-deficient mice*	Class-level support that fostamatinib corrects SYK-driven B-cell dysfunction/autoimmunity.
40903253	*Future directions in warm AIHA management*	Fostamatinib as an established, efficacious SYK inhibitor in immune disease.
40962713	*Fostamatinib and acute aortic dissection in ITP*	Safety caveat for chronic SYK inhibition.
31356155	*Chemical-genetics SYK mouse model*	Complementary tool confirming SYK-dependent BCR signaling.
25682127	*SYK molecular evolution in lamprey*	Evolutionary conservation of SYK tandem-SH2/kinase architecture.

Evidence-source types: **Human clinical** (P 33782605 case series; IUIS classification 41608114); **model organism** (33782605 mouse; 42026764; 31356155); **in vitro/structural** (23154170; 35398488); **lymphoma clinical correlation** (38503806).

Limitations and Knowledge Gaps

1. **Tiny cohort (n≈6):** All core clinical claims rest on a single 2021 case series. Penetrance, expressivity, sex ratio, natural history, and quantitative phenotype frequencies cannot be reliably estimated.
2. **No epidemiology:** Prevalence/incidence are unknown; the disease is ultra-rare with no registry data.
3. **Branch B is partly inferred:** The lymphoma-predisposition mechanism is supported by strong SYK/BCR biology (PMID: 35398488; PMID: 38503806) but has not been dissected specifically in IMD82 patient B cells at scale.
4. **Variant spectrum incomplete:** Only a few variants (notably p.Ser550Tyr) are characterized; the full allelic series and genotype–phenotype correlations are undefined.
5. **Therapeutic evidence is preclinical:** SYK-inhibitor/BMT rescue is demonstrated in mice; human treatment outcomes for IMD82 specifically are anecdotal, and fostamatinib carries vascular/thrombotic safety concerns (PMID: 40962713).
6. **No omics depth:** Patient-level transcriptomic, proteomic, metabolomic, or single-cell data for IMD82 are not available in the reviewed literature.
7. **Missing ontology/registry codes:** Orphanet/ICD/MeSH-specific mappings are not firmly established for this recently defined entity.

Proposed Follow-up Experiments / Actions

1. **Establish an international IMD82 patient registry** with genotype, functional pSYK data, longitudinal inflammation scores, and lymphoma incidence to define natural history, penetrance, and expressivity.

2. **Expand and functionally classify the *SYK* variant allelic series** (deep mutational scanning across the tSH2/interdomain-A/kinase regions) to build a genotype–phenotype and pSYK-activity map for variant interpretation (ACMG PS3 functional evidence).
3. **Single-cell multi-omics of patient PBMC/gut/lymphoma tissue** to resolve which cell types (B cells vs myeloid) drive Branch A vs Branch B and to identify predictive biomarkers.
4. **Prospective, biomarker-guided fostamatinib trial in IMD82** (with cardiovascular safety monitoring per PMID: 40962713), including pSYK/PI3K-AKT pharmacodynamic readouts.
5. **Define HSCT indications and outcomes** for severe/lymphoma-associated IMD82, leveraging the mouse WT-BMT rescue as rationale.
6. **Test whether early SYK inhibition reduces lymphoma incidence** in the Syk-Ser544Tyr knock-in model (chemoprevention proof-of-concept).
7. **Formalize ontology/registry entries** (Orphanet, ICD-11, MeSH) and curate HPO frequency annotations as new cases accrue.

Report compiled from 6 confirmed findings and 32 reviewed papers over 5 investigation iterations. Evidence source types are distinguished throughout as human clinical, model organism, in vitro/structural, or computational.
